Machine Learning in Bioinformatics

Projects

Group 3 & 4

AGH University of Krakow
Faculty of Computer Science

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Projects

Protein structure prediction (PSP)

Create an end-to-end framework that predicts 3D protein structure (in the form of a PDB file) from an amino acid protein sequence.

Protein function prediction (PFP)

Create an end-to-end framework that predicts protein function (in the form of Gene Ontology term probabilities) from an amino acid protein sequence.

Meeting schedule

- **Meeting 1:** overview of the projects; team assignment
- **Meeting 2:** train / test dataset preparation — **teamwork + progress report (5 min.)**
  - **PSP:** PDB and/or AFDB proteins
  - **PFP:** PDB/SIFTS or AFDB/GOA
- **Meeting 3:** end-to-end skeleton with working input/output — **teamwork + progress report (5 min.)**
- **Meeting 4:** adding evaluation framework — **teamwork + progress report (5 min.)**
  - **PSP:** RMSD, TM-score, GDT_TS, LDDT
  - **PFP:** CAFA-evaluator
- **Meeting 5:** adding baseline methods — **teamwork + progress report (5 min.)**
  - **PSP:** ESM3, ESMFold2 etc.
  - **PFP:** deepFRI, eggNOG-mapper etc.
- **Meeting 6:** model implementation — **teamwork + progress report (5 min.)**
- **Meeting 7:** final presentation (20 min.)

Team assignment

Group 3:

PSP team 1: 3 students
PSP team 2: 3 students

PFP team 1: 3 students
PFP team 2: 3 students

Group 4:

PSP team 1: 3 students
PSP team 2: 3 students

PFP team 1: 3 students
PFP team 2: 2 students